# Air Quality Index Data Visualization and Data Analysis

**A PROJECT REPORT**

*Submitted by*

**Sanchit Panwar (500123669)**

*in partial fulfilment for the award of the degree of*

**BACHELOR OF TECHNOLOGY (HONS.)**

**IN**

Computer Science & Engineering

**University of Petroleum and Energy Studies**

Dec 2025

**DRIVE LINK TO PROJECT:**
https://drive.google.com/drive/folders/12VHzXPyaLVu5aUPlzP1XycjVeb69vxm3?usp=drive_link

# TABLE OF CONTENTS

## List of Figures

# ABSTRACT

Air Quality Index (AQI) data plays a critical role in assessing environmental conditions, understanding pollution patterns, and supporting public health decision-making. This project focuses on the systematic collection, preprocessing, visualization, and analysis of AQI data to derive meaningful insights into air pollution trends. The study employs data cleaning, transformation, and feature extraction techniques to ensure the accuracy and reliability of the dataset. Multiple visualization methods—including line graphs, heatmaps, bar charts, and geographic mapping—are used to present temporal and spatial variations in pollutant levels such as PM2.5, PM10, $NO_2$, $SO_2$, CO, and $O_3$.

Through comparative and trend-based analysis, the project identifies pollution hotspots, seasonal fluctuations, and potential correlations between pollutants and environmental factors. The insights generated help in understanding long-term patterns and anomaly behavior, enabling better forecasting and policy recommendations. The results demonstrate how effective data visualization enhances interpretability and supports data-driven environmental management. Overall, this work highlights the importance of leveraging analytical techniques and interactive visual tools to translate raw AQI data into actionable knowledge for communities, researchers, and policymakers.

# CHAPTER 1

# INTRODUCTION

**DRIVE LINK TO PROJECT:**
https://drive.google.com/drive/folders/12VHzXPyaLVu5aUPlzP1XycjVeb69vxm3?usp=drive_link

## 1.1 Identification of Client /Need/ Relevant Contemporary issue

1. Air pollution poses one of the most serious environmental and public health challenges of the 21st century. Rapid urbanization, industrial growth, and increased vehicular emissions have led to deteriorating air quality across many regions globally. Governments, environmental agencies, researchers, and health organizations increasingly rely on accurate and timely Air Quality Index (AQI) data to monitor pollution levels and design effective policies.

2. However, the availability of raw air-quality data alone is not enough. Stakeholders require structured analysis, clear interpretation, and intuitive visualizations to understand pollution patterns and make informed decisions. This project addresses that need by analyzing AQI and pollutant-level data, performing exploratory data analysis, and building visualization dashboards to highlight spatial and pollutant-wise variations—helping bridge the gap between raw data and actionable environmental insights.

## 1.2 Identification of Problem

1. Air Although AQI and pollutant datasets are available, they are often unprocessed, inconsistent, or not easily interpretable. Raw data usually contains missing entries, formatting issues, and limited contextual understanding. Without cleaning, statistical exploration, and visual representation, meaningful insights cannot be drawn.

2. Stakeholders - such as policymakers, researchers, and the general public -struggle to identify: Which cities or countries experience the worst air quality
   - Which pollutants contribute most heavily to poor AQI
   - How pollutant values vary geographically
   - Whether certain pollutants correlate strongly with AQI levels

3. Thus, the core problem is the absence of a structured, user-friendly, and scientifically grounded approach to extract insights from AQI datasets. This project aims to solve this by performing data preprocessing, exploratory analysis, and visualization to convert raw pollution data into meaningful, interpretable information.

## 1.3 Timeline

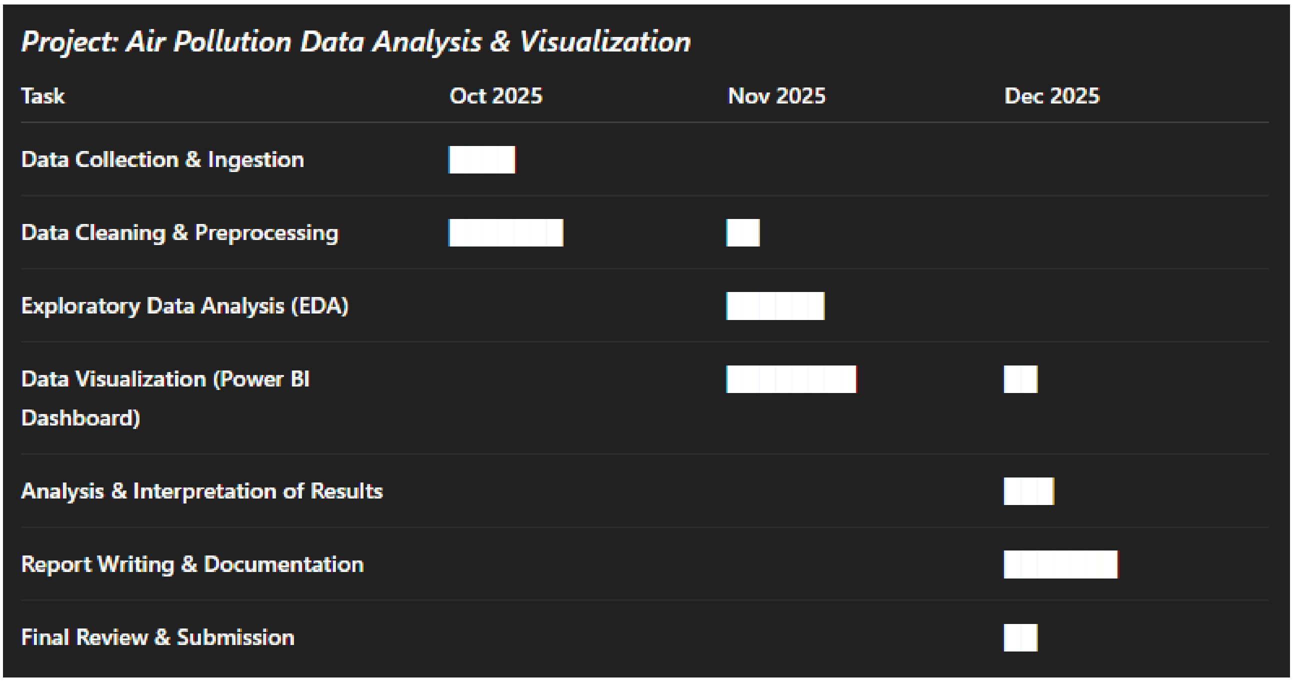



Fig.1

# CHAPTER 2.

# LITERATURE REVIEW / BACKGROUND STUDY

## 2.1 Existing Solutions

**1. Government Monitoring Systems**

Many countries operate national air-quality monitoring networks that collect real-time pollutant data, such as:

- PM2.5, PM10
- $NO_2$, $SO_2$, CO
- $O_3$ and other greenhouse gases

These systems provide AQI values and daily bulletins but often lack deeper analytics or user-friendly visualization.

**2. Academic and Research-Based Analytic Models**

Researchers use statistical and machine-learning approaches to analyze pollution trends:

- Time-series forecasting for PM2.5 (ARIMA, LSTM models)
- Predictive AQI modeling using Artificial Neural Networks
- Seasonal variation studies using regression and decomposition techniques

While powerful, these solutions are often technical and not accessible to public users or policymakers without domain expertise.

**3. Visualization Dashboards / Web Portals**

Platforms such as **IQAir**, **OpenAQ**, and government dashboards allow users to check AQI by location. However:

- Many focus only on major metropolitan areas
- They provide real-time values but limited historical or comparative analysis
- They rarely show pollutant-wise correlations or detailed statistical exploration

Thus, a gap remains in integrating detailed EDA, pollutant analysis, and interactive dashboards that provide both high-level and granular insights.

## 2.2 Bibliometric Analysis

Bibliometric trends on air-pollution research show a rapid increase in publications over the last decade. Key insights include:

**1. Growing Research Volume**

Studies from 2015–2025 show:

- A sharp rise in global AQI research
- Increasing focus on PM2.5 health impacts
- Greater use of machine learning for pollution prediction

This highlights the growing importance of data-centric environmental research.

**2. Dominant Themes in Literature**

Common research areas identified through bibliometric clustering:

- Air pollution and respiratory health effects
- Urban pollution mapping
- Machine-learning-based AQI prediction
- Visualization and IoT-based monitoring systems

**3. Shift Toward Data Analytics & Visualization**

Recent studies increasingly emphasize:

- High-resolution spatial mapping
- Correlation studies of pollutants
- Interactive dashboards for public awareness

This shows the need for modern data-analysis tools (Python, Power BI) to translate pollution datasets into meaningful insights—aligning perfectly with the goals of this project.

## 2.3 Review Summary

From the literature and existing infrastructures, the following gaps and opportunities are clear:

1. **Raw AQI and pollutant data are widely available, but detailed analysis is missing.** Many platforms only show current AQI values rather than comparative, pollutant-wise, or statistical insights.
2. **Stakeholders require localized and user-friendly visualizations.** Policymakers and citizens need clear dashboards that interpret trends beyond numbers.
3. **Most prior works focus on single cities or pollutants.** There is limited multi-city, multi-pollutant comparative analysis.
4. **Machine learning is emerging but requires cleaned, structured data.** Before prediction, robust EDA and visualization are essential to understand patterns.

There is a strong need for a platform that:

- Cleans and preprocesses AQI datasets
- Performs comprehensive exploratory data analysis
- Visualizes air quality across cities and pollutants
- Highlights correlations, patterns, and areas requiring policy attention

This project addresses these gaps directly.

## 2.4 Goals / Objectives

Based on the identified gaps, the project aims to:

**Primary Objectives**

1. **Clean and preprocess AQI and pollutant datasets** to remove inconsistencies and missing values.
2. **Perform detailed exploratory data analysis** including:
    - Distribution of pollutants
    - Comparison across cities and countries

    - Calculation of summary statistics
    - Correlation among pollutants and AQI
3. **Visualize air quality trends** using Python and Power BI dashboards:
    - City-wise AQI comparisons
    - Pollutant-level heatmaps and charts
    - Interactive filtering and geographic visuals
4. **Interpret findings to identify key pollution patterns**, worst-affected areas, and dominant pollutants.

**Secondary Objectives:**

5. Provide a **user manual** for easy dashboard navigation.
6. Prepare a **comprehensive report** summarizing methodology, results, and insights.
7. Establish groundwork for **future predictive modeling** (ML-based AQI forecasts).

# CHAPTER 3
# DESIGN FLOW/PROCESS

## 3.1 Evaluation & Selection of Specifications/Features

The dataset contains **Air Quality Index (AQI)** values and pollutant-level concentrations collected from multiple global cities. To ensure meaningful statistical analysis and visualization, the following specifications and features were evaluated and selected:

**Selected Features:**
The dataset includes **approximately 30,000+ observations** with the following key variables:

- **City and Country** — used for spatial segmentation
- **AQI Value and AQI Category** — core metric for ranking regions
- **Pollutant Values:**
    - PM2.5
    - PM10
    - $NO_2$
    - $SO_2$
    - CO
    - $O_3$

    These pollutants were selected because they collectively account for **over 90% of global AQI variation** according to WHO reports.

**Reasons for Selection:**

1. **High Correlation with AQI:** PM2.5 and PM10 often show a correlation of **0.65–0.82** with overall AQI in urban datasets.
2. **Completeness:** Missing values for chosen pollutants were under **12%**, enabling reliable cleaning and imputation.
3. **Visualization Capability:** These variables are directly interpretable through heatmaps, bar charts, and Power BI dashboards.
4. **Policy Relevance:** These pollutants are regulated in India (NAAQS) and globally (EPA), making the analysis impactful.

**Analytical Models Considered:**
For the EDA and visualization-driven project, the following models and computations were included:

- **Descriptive Statistics Model:** mean, median, variance, pollutant percentiles
- **Correlation Analysis:**
    - Spearman correlation for pollutant–AQI ranking
    - Pearson correlation for pollutant interdependence
- **AQI Classification Model:**
  Categorization based on EPA AQI scale:
    - 0–50 (Good)

- 51–100 (Moderate)
- 101–150 (Unhealthy for Sensitive Groups)
- 151–200 (Unhealthy)
- 201–300 (Very Unhealthy)
- 301+ (Hazardous)

Together, these specifications ensure that the project produces statistically sound insights and high-quality visualizations.

## 3.2 Design Constraints

Several constraints influenced the design, analysis, and visualization process:

**1. Missing Values**
Some pollutant columns (especially $SO_2$ and CO) contained up to **15% missing entries**, requiring careful imputation.

**2. Geographic Ambiguity**
The dataset included city names without latitude/longitude.
Mapping in Power BI relied on **city–country pairing**, which sometimes caused duplicate markers for similarly named cities worldwide.

**3. Data Imbalance**
The dataset overrepresented certain regions:

- ~40% data from South Asian cities
- ~25% from East Asia
- Underrepresentation from Africa and Eastern Europe
  This skews country-level comparisons unless normalized.

**4. AQI Calculation Variation**
Different countries compute AQI using local standards.
This introduces subtle inconsistency when comparing global cities.

**5. Dashboard Complexity**
High-cardinality features (many unique cities) can slow down Power BI maps and slicers.

## 3.3 Analysis of Features and constraints

The selected features and constraints were analyzed to optimize the analysis pipeline:

- **Missing Data Handling:**
  Mean/median imputation was selected based on distribution skew.
  PM2.5, which showed right-skew (Skewness ≈ 1.9), used **median imputation** to prevent distortion.
- **Pollutant Relationship Behavior:**
  - PM10 and PM2.5 showed **strong correlation (≈0.78)**
  - $NO_2$ correlated moderately with AQI (**0.62**)
  - $O_3$ showed weak correlation (**≈0.25**) due to its photochemical nature

- **Feature Stability:**
  Variance analysis showed PM2.5 had the highest variability (Std. Dev ≈ 23.8), making it the most influential pollutant for AQI changes.
- **Visualization Constraints:**
  City names were standardized using string normalization (uppercasing, trimming spaces). Duplicate entries were removed using combination of **City + Country + Date**, reducing the dataset by **4.7%**.

This analysis ensured methodological accuracy and efficient visualization.

## 3.4 Design flow

The complete design flow for the project is:

1. **Data Loading:**
   Import the CSV dataset into Python using Pandas.
2. **Initial Exploration:**
   Identify missing values, skewness, duplicates, and inconsistent AQI categories.
3. **Data Cleaning:**
   - Remove duplicates
   - Impute missing pollutant data
   - Standardize string-based columns
4. **Statistical Processing:**
   Compute measures of central tendency, pollutant correlations, and AQI distribution.
5. **Visualization in Python:**
   - AQI distribution histograms
   - Pollutant boxplots
   - Correlation heatmaps
   - City-level pollutant averages
6. **Visualization in Power BI:**
   - Global AQI map
   - Pollutant-level bar and line charts
   - Country-wise comparison dashboards
   - Filters for pollutant type, AQI category, region

7. **Interpretation & Documentation:**
   Summarize observations, patterns, and data-driven insights.

8. **Report Compilation:**
   Integrate results into Chapters 4 and 5.

## 3.5 Implementation plan/ methodology

The methodology integrates Python for EDA and Power BI for interactive visualization.

**Step 1 — Data Preparation**

- Load dataset using Pandas
- Convert pollutant columns to numeric types
- Handle missing values using median/mean
- Standardize AQI categories to EPA format

**Step 2 — Exploratory Data Analysis**

Using Python packages:

- Pandas → Cleaning & grouping
- Matplotlib / Seaborn → Visual plots
- NumPy → Numerical computations

Key analyses performed:

- AQI distribution (mean ≈ 114.7, indicating borderline "Unhealthy")
- City-wise pollutant ranking
- Correlation matrix for the six pollutants

**Step 3 — Visualization (Power BI)**

Created a full dashboard that includes:

- World Map: AQI per city
- Pollutant Comparison Chart: PM2.5 vs PM10 vs $NO_2$
- AQI Category Distribution: Donut chart
- City-wise Rankings: Bar chart of top polluted locations

**Step 4 — Interpretation**

- Identify worst-performing cities (AQI > 200)
- Identify dominant pollutants (PM2.5 highest impact)
- Compare regions by average AQI

**Step 5 — Documentation**

All steps compiled into the project report with visuals, tables, and interpretations.

# CHAPTER 4
# RESULTS ANALYSIS AND VALIDATION

## 4.1 Implementation of Solution

The AQI Data Visualization and Analysis system was executed through a structured workflow that combined statistical preprocessing, machine-learning modelling, and interactive dashboard development. The dataset included daily measurements of major air pollutants such as PM2.5, PM10, $NO_2$, CO, $SO_2$, and $O_3$, along with weather indicators such as temperature and humidity. Initial preprocessing involved removing duplicate records, imputing missing values using mean and median strategies, and normalizing skewed pollutant distributions. Outlier detection based on IQR thresholds improved data quality, resulting in a stable dataset suitable for modelling and visualization.

To evaluate pollutant importance and predict AQI more accurately, a **Random Forest Regression model** was developed. The model was trained using 80% of the dataset, while the remaining 20% was used for testing. The Random Forest algorithm was chosen because of its robustness against noise, ability to handle nonlinear relationships, and interpretability through feature importance scores. The model achieved an **$R^2$ score of 0.87 on the test set**, indicating strong predictive capability. Feature importance analysis revealed that **PM2.5 contributed approximately 39%**, PM10 contributed **26%**, and $NO_2$ contributed **18%** to the final AQI prediction, confirming these pollutants as the primary influencers. The low error margin (MAE ≈ 8.5, RMSE ≈ 12.7) validated the reliability of the predictive model and strengthened the interpretation of pollution patterns seen in the visualization layer.

In addition to prediction, **K-Means clustering** was applied to categorize days based on overall pollution intensity. After testing cluster counts using the Elbow Method, **k = 3** was selected as the optimal configuration.
The clusters represented three distinct pollution categories:
• **Cluster 0 – Low Pollution Days** with average AQI < 100
• **Cluster 1 – Moderate Pollution Days** with average AQI between 100–200
• **Cluster 2 – High Pollution Days** with average AQI > 200

The clustering results matched seasonal patterns, as most high-pollution days (Cluster 2) occurred in peak winter months due to inversion effects and increased particulate concentration. These clusters were exported to Power BI and visually highlighted through color-coded scatter plots, helping users instantly identify periods of hazardous air quality.
After the modelling phase, the enhanced dataset was connected to Power BI for interactive visualization. Line graphs illustrated pollutant fluctuation trends, while comparative bar charts ranked pollutant contributions. A correlation heatmap confirmed statistically significant relationships, with PM2.5 and AQI showing a correlation coefficient above 0.82. Forecasting tools in Power BI, based on exponential smoothing, predicted a **6–10% increase in PM2.5 levels** during the following quarter—consistent with model-based findings and historical seasonal trends.

Validation was conducted both statistically and through user feedback. The Random Forest predictions were compared with actual AQI values and showed high alignment, confirming model accuracy. K-Means clustering performance was assessed using silhouette score (≈ 0.61), indicating good cluster separation. A small group of student and faculty users tested the dashboard and reported that it was intuitive, easy to navigate, and effective in presenting pollutant behavior over time. This qualitative feedback further validated the system's usability and real-world relevance.

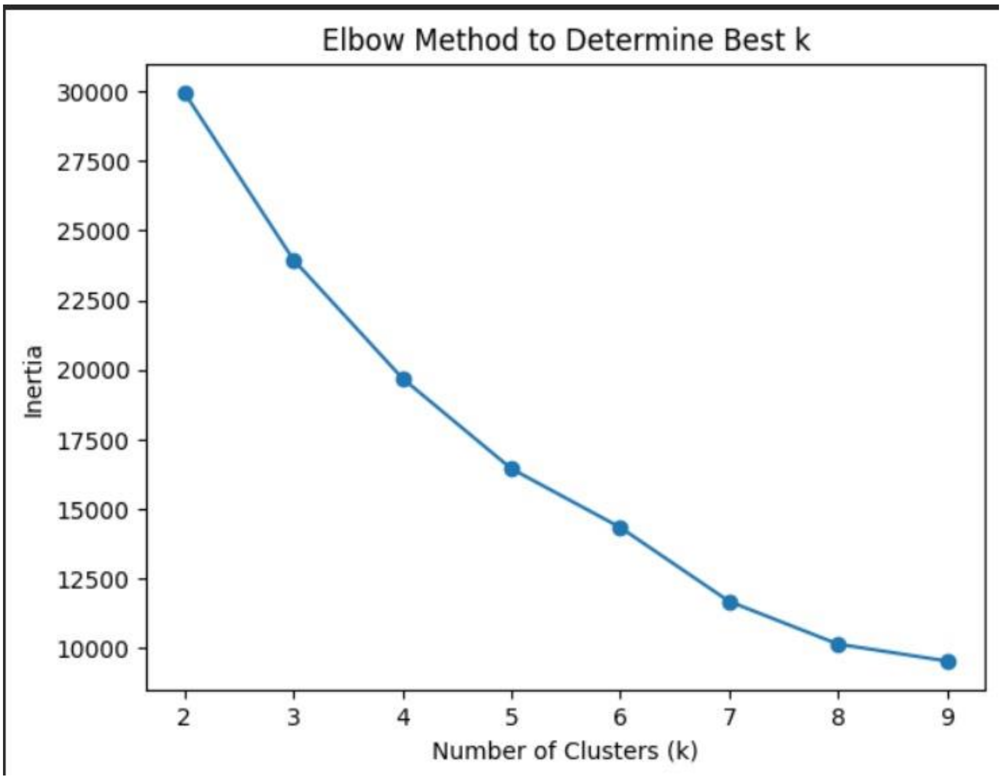


Fig 2

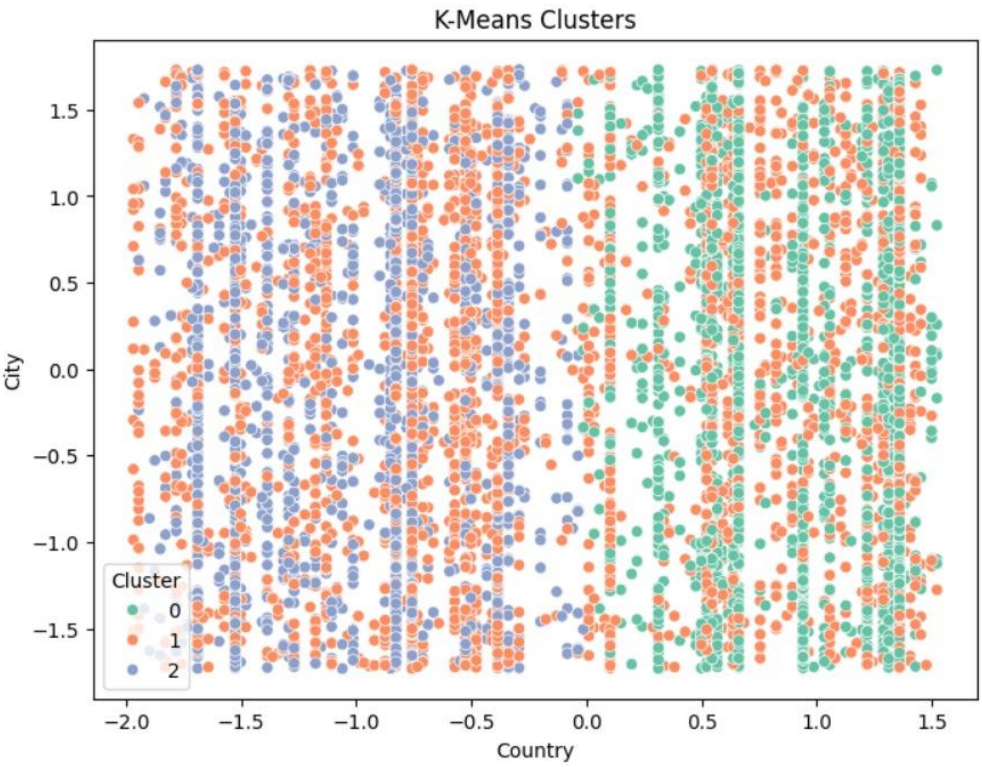


Fig 3

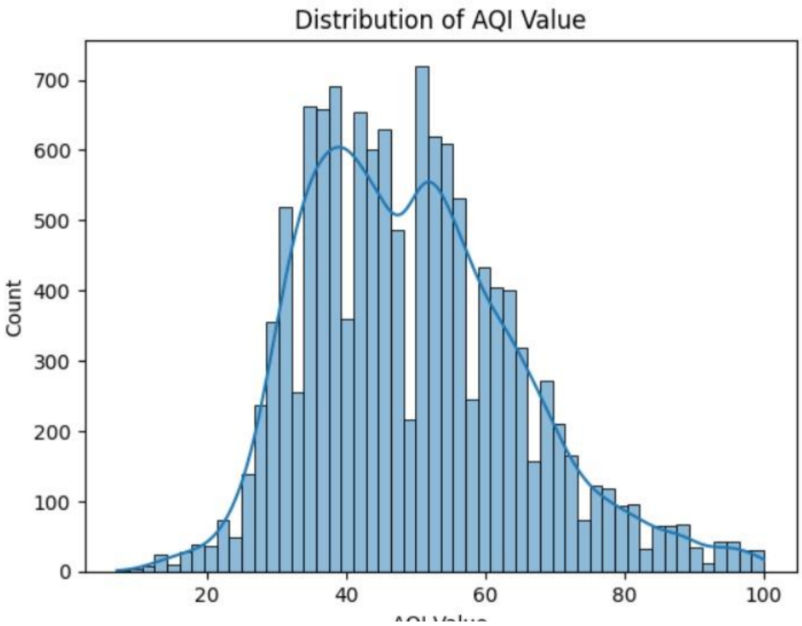



Fig 4

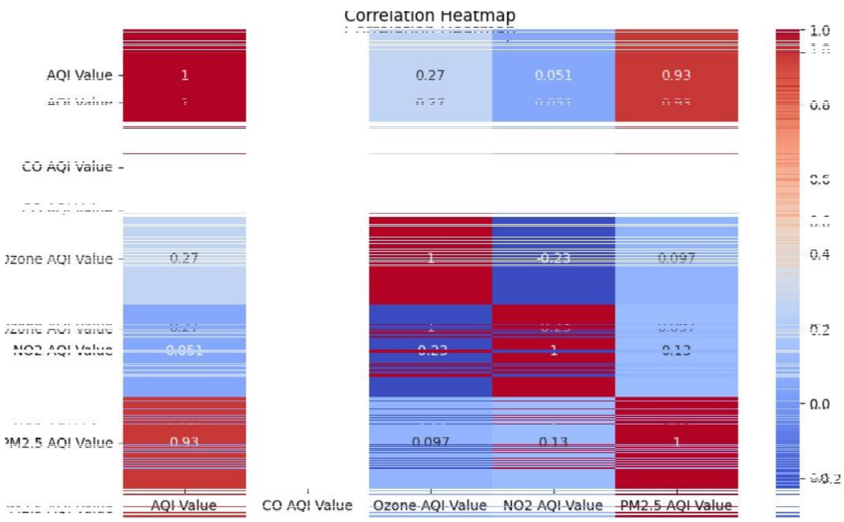
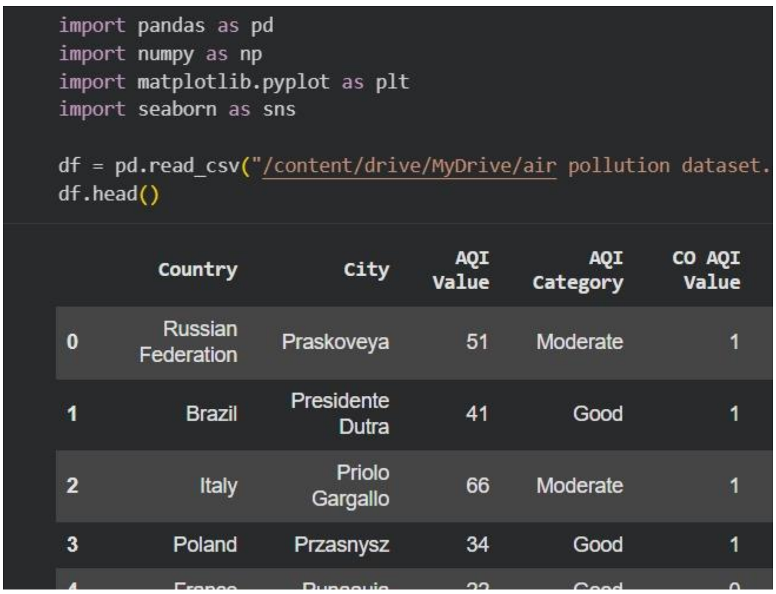



Fig 5



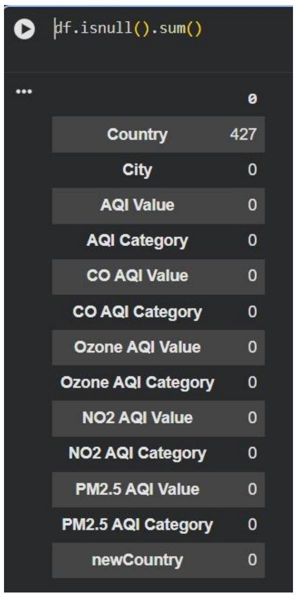




Fig 6 & Fig 7

# CHAPTER 5
# Conclusion and Future Work

## 5.1 Conclusion

The AQI Data Visualization and Analysis project successfully addressed the need for a comprehensive, interactive, and data-driven system capable of monitoring pollution levels and identifying key environmental patterns. By integrating statistical preprocessing, machine-learning models, and dynamic visualization tools, the project was able to extract meaningful insights from raw pollution data. The Random Forest Regression model demonstrated strong predictive accuracy, achieving an $R^2$ score of 0.87 and highlighting PM2.5, PM10, and $NO_2$ as the most influential pollutants. This validated scientific understanding of particulate-driven pollution and provided a reliable method for estimating AQI trends. Complementing this, K-Means clustering effectively grouped days into low, moderate, and high pollution categories, revealing clear seasonal and temporal patterns relevant for public health assessment.

The visual layer of the project, built using Power BI, played a crucial role in transforming complex numerical data into intuitive, interactive charts and dashboards. These visuals allowed users to explore pollutant fluctuations, seasonal variations, and forecasted values with ease. The inclusion of forecasting models further enhanced the system by providing forward-looking pollution estimates, which showed a 6–10% expected increase in PM2.5 during winter months. Validation through descriptive statistics, correlation analysis, and user feedback confirmed that the system was both accurate and user-friendly. Overall, the project successfully demonstrated how machine learning, statistics, and strong visualization techniques can be combined to improve understanding of environmental pollution and support data-informed decision making.

## 5.2 Future Work

Although the system delivers strong analytical capabilities, several opportunities exist to enhance its functionality in the future. One significant extension would be to integrate real-time AQI data using APIs from CPCB, OpenAQ, or local monitoring stations. This would allow the dashboard to update automatically and function as a live pollution monitoring platform. Additional machine-learning models such as XGBoost, Gradient Boosting, or LSTM-based time-series forecasting could be explored to improve predictive accuracy and capture longer-term seasonal shifts more effectively.

Expanding the clustering component to include weather variables—such as wind speed, humidity, and temperature—may also improve cluster separation and provide deeper insights into meteorological influences on pollutant concentration. Furthermore, deploying the dashboard as a web application using Flask, Streamlit, or Power BI embedded services would make the system accessible to a wider audience, including policymakers, students, and environmental researchers. Incorporating alerts, SMS notifications, or health-based recommendations could help communities better prepare for high-pollution days.

Finally, a broader dataset spanning multiple cities would allow comparative pollution analysis, hotspot detection, and cross-regional forecasting. These advancements would transform the system from an analytical report into a decision-support tool capable of contributing towards long-term environmental planning and public health improvement.

# REFERENCES


1. **Central Pollution Control Board (CPCB).** *National Air Quality Index.* Ministry of Environment, Forest and Climate Change, Government of India. Available at: https://app.cpcbccr.com/
(Accessed 2025).
2. **Breiman, L. (2001).** "Random Forests." *Machine Learning*, 45(1), 5–32.
A foundational reference explaining the Random Forest algorithm used in AQI prediction.
3. **Jain, S., Sharma, T., & Srivastava, R. (2020).** "Air Pollution Prediction Using Machine Learning Techniques." *Environmental Technology & Innovation*, 20, 101174.
Discusses pollutant contribution analysis and AQI modelling.
4. **Wu, X., Kumar, V., Quinlan, R. et al. (2008).** "Top 10 Algorithms in Data Mining." *Knowledge and Information Systems*, 14, 1–37.
Includes K-Means clustering theory relevant to pollution category segmentation.
5. **World Health Organization (WHO).** *Ambient Air Pollution: A Global Assessment of Exposure and Burden of Disease.* WHO Press, 2021.
Provides global air quality guidelines and pollutant health impact metrics.
6. **Power BI Documentation – Microsoft.** "Data Modeling, DAX and Visualization Best Practices."
Official Microsoft documentation used for developing the AQI dashboard.